

Faculty of Information Technology and Artificial Intelligence

Introduction to System Administration

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1. Objectives

The objective of this assignment is to use an AI agent inside an Ubuntu virtual machine to generate and execute Linux commands automatically from natural-language prompts. The assignment demonstrates how an AI agent can interpret a request, select a suitable command, and run it in the terminal after user approval.

2. Experimental Environment and Materials

The experiments were performed on Ubuntu Linux running inside a VMware virtual machine, using the Ubuntu Terminal (CLI). Google Antigravity CLI was used as the AI agent, powered by Google Gemini AI, and it was authenticated through Google OAuth. The agent was used for autonomous Linux command generation and execution.

3. Procedure

3.1 Installing Google Antigravity CLI

Google Antigravity CLI was installed inside the Ubuntu virtual machine using the official installation script.

Command(s):

```
curl -fsSL https://antigravity.google/cli/install.sh | bash
```

This command downloads the installer script and executes it through bash, installing the AI agent on the system.

3.2 Preparing the Working Directory

A dedicated directory was created for this assignment. The following commands were entered manually by the student.

Command(s):

```
mkdir -p ~/AI_Assignment2
cd ~/AI_Assignment2
~/.local/bin/agy
```

These commands create the working directory, change into it, and launch the AI agent (agy).

3.3 Google Authentication

Google OAuth was selected as the login method, and the AI agent was authenticated using a Google account. After successful authentication, Antigravity CLI was ready to receive natural-language instructions and execute Linux commands.

3.4 Experiment 1 – Displaying Network Interfaces

In this experiment, the AI agent was instructed to identify the network interfaces available on the Ubuntu machine.

AI Prompt:

```
"Show all network interfaces on my Ubuntu machine using ip. Generate
and execute the command yourself, and display the actual terminal
output. Keep your response short."
```

AI-Generated Command:

```
ip link show
```

The AI agent independently selected the command and, as shown in Figure 1, requested permission before running it.

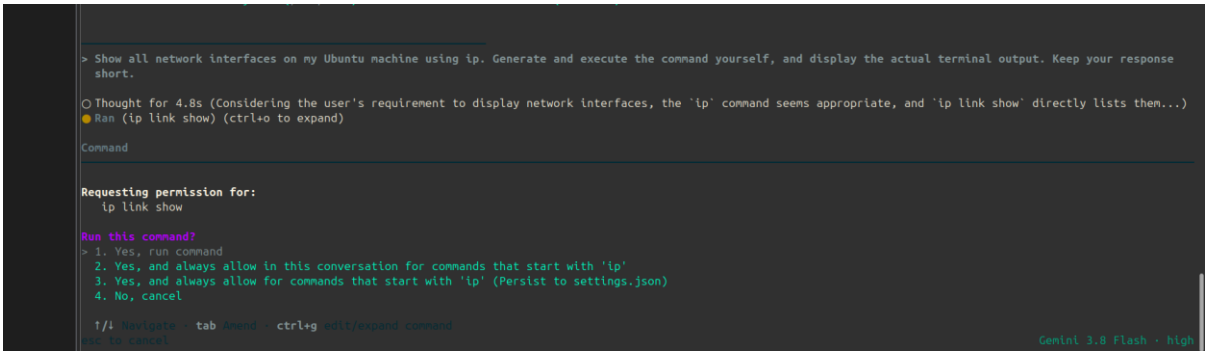


Figure 1: The AI agent requesting permission to execute ip link show

After approval, the command was executed and the output was displayed, as shown in Figure 2.

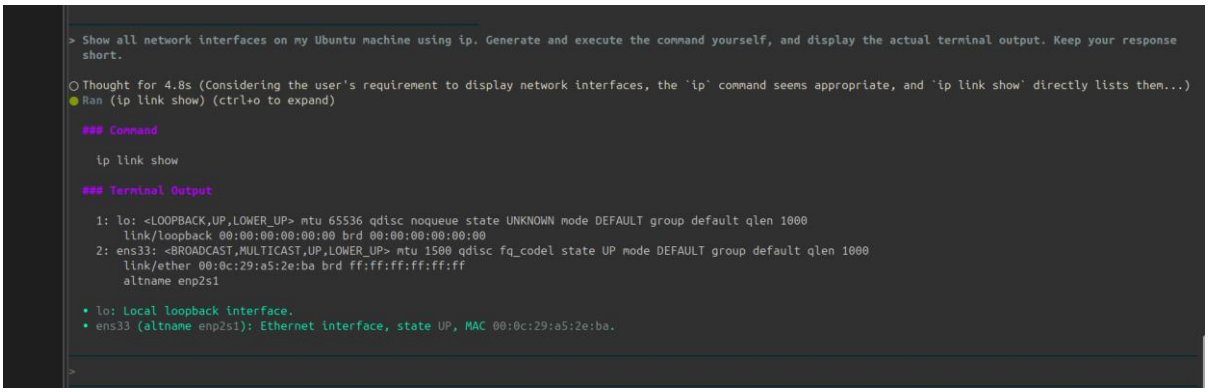


Figure 2: Execution results of ip link show

Result:

The AI agent successfully executed ip link show and displayed the available network interfaces, including the loopback interface (lo) and Ethernet interface (ens33), which was in the UP state.

3.5 Experiment 2 – Checking Listening TCP Ports

In this experiment, the AI agent was asked to identify the TCP ports currently listening on the Ubuntu machine.

AI Prompt:

```
"Check which TCP ports are currently listening on my Ubuntu machine.
Choose and execute the appropriate Linux command yourself. Display
the real terminal output and provide a very short explanation. Do not
modify anything."
```

AI-Generated Command:

```
ss -tlnp
```

The options used by the AI agent have the following meanings:

- `-t`: Display TCP sockets.
- `-l`: Display listening sockets.
- `-n`: Show numeric addresses and ports.
- `-p`: Display associated processes when available.

As shown in Figure 3, the AI agent requested permission before executing the command.

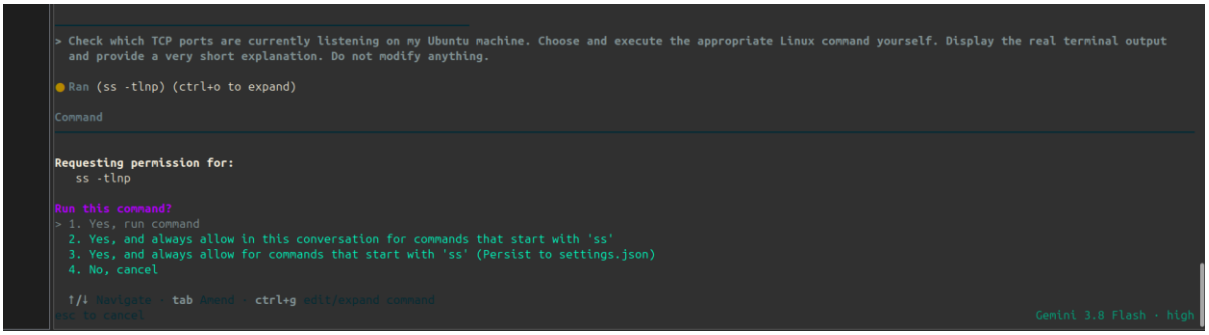


Figure 3: The AI agent requesting permission to execute ss -tlnp

After approval, the command was executed and the listening TCP ports were displayed, as shown in Figure 4.

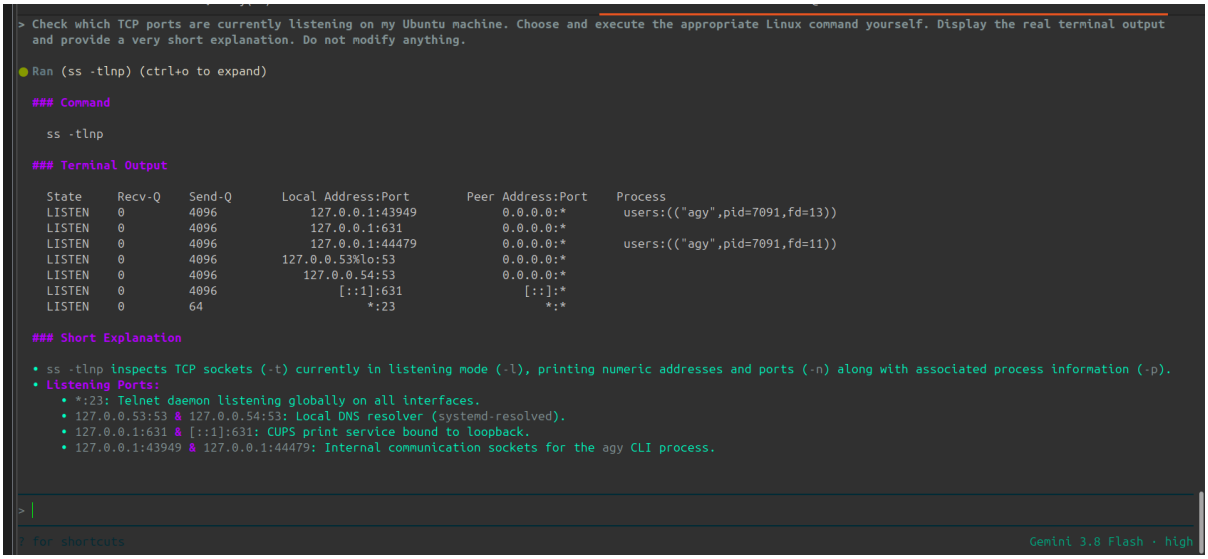


Figure 4: Output of ss -tlnp showing the listening TCP ports

Result:

The AI agent successfully executed ss -tlnp and displayed the TCP ports listening on the Ubuntu system. The results included local services and port 23 listening on all interfaces, which is commonly associated with Telnet.

4. Conclusions

Google Antigravity CLI successfully generated and executed Linux commands based on natural-language requests inside the Ubuntu virtual machine. The two experiments demonstrated AI-assisted network interface inspection using ip link show and listening TCP port identification using ss -tlnp. In both cases, the commands were selected by the AI agent and executed only after the user approved them. This shows that AI agents can simplify routine system administration tasks while keeping the user in control of command execution.

5. References

[1] Google, "Google Antigravity CLI Documentation," https://antigravity.google.

[2] Linux man-pages project, "ip-link(8) – network device configuration," https://man7.org/linux/man-pages/man8/ip-link.8.html.

[3] Linux man-pages project, "ss(8) – another utility to investigate sockets," https://man7.org/linux/man-pages/man8/ss.8.html.